Year 13 Maths - Further differentiation

Similar questions - Product, quotient and chain rules

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Differentiate with respect to x, simplifying where possible:

(a) x^2e^{-3x} **[3]**

(b) $\dfrac{2x+1}{x^2+4}$ **[3]**

(c) $\tan\left(3x^2\right)$ **[2]**

(d) Hence find the values of x at which the curve $y = x^2e^{-3x}$ has stationary points. **[2]**

2 A curve has equation $y = \dfrac{\ln x}{x^2}$ for $x > 0$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{1-2\ln x}{x^3}$. **[3]**

(b) Find the exact coordinates of the stationary point. **[3]**

(c) Determine the nature of the stationary point by considering the sign of $\dfrac{dy}{dx}$ on either side of it. **[2]**

Question 1

(a) Product rule: $2xe^{-3x} + x^2(-3e^{-3x}) = xe^{-3x}(2-3x)$

(b) Quotient rule: $\dfrac{2(x^2+4)-(2x+1)(2x)}{(x^2+4)^2} = \dfrac{8-2x-2x^2}{(x^2+4)^2}$

(c) Chain rule: $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan u = \sec^2 u \dfrac{\mathrm{d}u}{\mathrm{d}x}$ gives $6x\sec^2(3x^2)$

(d) $xe^{-3x}(2-3x) = 0$ gives $x = 0$ or $x = \frac{2}{3}$

Question 2

(a) Quotient rule: $\dfrac{\frac{1}{x}\cdot x^2 - \ln x \cdot 2x}{x^4} = \dfrac{x-2x\ln x}{x^4} = \dfrac{1-2\ln x}{x^3}$

(b) $1-2\ln x = 0$ gives $\ln x = \frac{1}{2}$, $x = \sqrt{e}$. $y = \dfrac{1/2}{e} = \dfrac{1}{2e}$. Point $\left(\sqrt{e}, \dfrac{1}{2e}\right)$

(c) For $x < \sqrt{e}$, $\ln x < \frac{1}{2}$ so $\dfrac{\mathrm{d}y}{\mathrm{d}x} > 0$; for $x > \sqrt{e}$ it is negative. The gradient changes from positive to negative, so it is a maximum